

from iteration 1.

$$m = \begin{bmatrix} 1.45 \\ 0.87 \end{bmatrix}$$

$$b = 0.88$$

$$\alpha = 0.01$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\hat{y} = X_m + b$$

$$X_m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + b$$

x_m

$$\begin{aligned} & (1)(-1.45) + (3)(0.87) \\ &= -1.45 + 2.61 \\ &= 1.16 \end{aligned}$$

$$\begin{aligned} & (4)(-1.45) + (10)(0.87) \\ &= -5.80 + 8.70 \\ &= 2.90 \end{aligned}$$

$$x_m = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

$$y = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$= \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

$$e = \hat{y} - y$$

$$= \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

Gradient for m

$$\nabla_m = \frac{2}{n} X^T e$$

$$n = 2$$

$$\frac{2}{n} = 1$$

$$\nabla_m = X^T e$$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

Multiply $\begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$

$$\begin{aligned} & (1)(-2.96) + (4)(-2.22) \\ &= -2.96 - 8.88 \\ &= -11.84 \end{aligned}$$

$$\begin{aligned} & (3)(-2.96) + (10)(-2.22) \\ &= -8.88 - 22.20 \\ &= -31.08 \end{aligned}$$

$$\nabla m = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

Gradient for b

$$V_b = \frac{2}{n} \sum e$$

$$\frac{2}{2} = 1$$

$$\begin{aligned} V_b &= -2.96 + (-2.22) \\ &= -5.18 \end{aligned}$$

Update m

$$\begin{aligned} m_{\text{new}} &= m - \alpha V_m \\ &= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix} \end{aligned}$$

$$0.01 \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix} = \begin{bmatrix} -0.1184 \\ -0.3108 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - \begin{bmatrix} -0.1184 \\ -0.3108 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

Update b

$$\begin{aligned} b_{\text{new}} &= b - \alpha \nabla b \\ &= 0.88 - 0.01 (-7.18) \\ &= 0.88 + 0.0718 \\ &= 0.9518 \end{aligned}$$

Updated parameters

$$m = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

$$b = 0.9518$$